Past Paper 2023 (i)

Differentiate between deterministic and stochastic model?

Unlike deterministic models that produce the same exact results for a particular set of inputs stochastic models are the opposite.

(ii)

Discuss Statistical properties of OLS estimators.

(iii)

Discuss causes of multicollinearity

1. Inaccurate use of different types of variables.
2. Poor selection of questions or null hypothesis.
3. The selection of a dependent variable.

4- Variable repetition in a linear regression model.

IV

Differentiate between time series and cross sectional data.

(V)

What is OLS?

OLS stands for Ordinary Least square its a statistical method used to estimate the relationship between a dependent variable and one or more independent variable.

(vi)

Discuss various functional forms of regression function?

The simplest functional form is the linear functional form

① Linear $Y = B_0 + B_1 X + \Sigma$
② Log Linear = $\ln(Y) = B_0 + B_1 X + \Sigma$
③ Log Log = $\ln Y = B_0 + B_1 \ln(X) + \Sigma$
④ Quadratic $Y = B_0 + B_1 X + \beta_2 X^2 + \Sigma$

where the relationship between the dependent variable and an independent variable is graphically represented by a straight line.

⑤ Exponential $Y = e^{B_0 + B_2 X + \Sigma}$.

Question: 2

The estimator model pertaining --- errors in parenthesis.

$CM = \overset{B_1}{263.6416} - \overset{B_2}{0.0056} PGNP_1 - \overset{\beta_3}{2.2316} FLR$

$Se = (11.5932)(0.0019)(0.2099)$

a Is coefficient of PGNP statistically significant

b Is coefficient of FLR statistically significant

c Are both coefficient Statistically coefficient.

i $\frac{0.0056}{0.0019} = -2.95$

Not significant

ii $\frac{-2.2316}{0.2099} = -10.63$

Not significant

iii Both coefficient statistcally not significant

(iii)

Obtain regression line (Y and x) interpret the results calculate standard error of coefficient and comment on statistical significance of X- variable

Y	2.75	2.15	4.40	5.50	3.20	4.30	2.31	4.30	3.70
X	29.5	26.3	32.2	36.5	27.2	27.5	28.3	30.3	28.5

Y	X	xy	x^2	Y^2
2.75	29.5	81.125	870.25	7.5625
2.15	26.3	56.545	691.69	4.6225
4.40	32.2	141.68	1036.84	19.36
5.50	36.5	200.75	1332.25	30.25
3.20	27.2	87.04	789.84	10.24
4.30	27.5	118.25	756.25	18.49
2.31	28.3	65.373	800.89	5.3361
4.30	30.3	130.29	918.09	18.49
3.70	28.5	105.45	812.25	13.69
32.61	266.3	986.503	7958.35	128.0411

$y = 32.61$, $x = 266.3$, $x^2 = 7958.35$

$xy = 986.503$, $y^2 = 128.0411$ $N = 9$

$$\beta_2 = \frac{n \Sigma xy - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$= \frac{9(986.503) - (266.3)(32.61)}{9(7958.35) - (266.3)^2}$$

$$= \frac{8878.527 - 8684.043}{71625.15 - 70915.69}$$

$$= \frac{194.484}{709.46}$$

$$\beta_2 = 0.2741$$

$$X = \frac{\Sigma x}{n} = \frac{266.3}{9} = 29.588$$

$$\bar{y} = \frac{\Sigma Y}{n} = \frac{32.61}{9} = 3.6233$$

$$\bar{\beta}_1 = \bar{Y} - \beta_2 \bar{X}$$

$$\beta_1 = 3.6233 - (0.274)(29.588)$$

$$\beta_1 = 3.6233 + (8.107112)$$

$$11.730412$$

$$Y = \beta_1 + \beta_2 X$$

$$Y = 11.730412 + 0.2741(29.5)$$

$$y = 11.730412 + 8.08595$$